

How Reliable Are VLM Judges? Conformal Prediction Reveals Task-Dependent Uncertainty in Multimodal Evaluation

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Abstract

Vision-language models (VLMs) are increasingly used as automated judges for multimodal systems, yet their scores provide no indication of reliability. We study this problem through conformal prediction, a distribution-free framework that converts a judge’s point score into a calibrated prediction interval using only score-token log-probabilities, with no retraining. We present the first systematic analysis of conformal prediction for VLM-as-a-Judge across 3 judges and 14 visual task categories. Our results show that evaluation uncertainty is strongly task-dependent: intervals cover $\sim 40\%$ of the score range for aesthetics and natural images but expand to $\sim 70\%$ for chart and mathematical reasoning, yielding a quantitative reliability map for multimodal evaluation. We further identify a failure mode not captured by standard evaluation metrics, *ranking-scoring decoupling*, where judges achieve high ranking correlation while producing wide, uninformative intervals, correctly ordering responses but failing to assign reliable absolute scores. Finally, we show that interval width is driven primarily by task difficulty and annotation quality, i.e., the same judge and method yield $4.5\times$ narrower intervals on a clean, multi-annotator captioning benchmark.

1 Introduction

Vision-Language Models (VLMs) are increasingly used as automated judges to evaluate multimodal AI systems (Chen et al., 2024; Li et al., 2025). This *VLM-as-a-Judge* paradigm extends LLM-based evaluation to tasks requiring visual understanding, including chart comprehension, visual question answering, image captioning, and AI-generated image quality. As multimodal agents are deployed in high-stakes settings such as medical imaging, web navigation, and document analysis, the reliability of these evaluations becomes critical.

However, VLM judges produce only point scores, offering no indication of reliability. A score of four out of five (4/5) may reflect confident correctness or substantial uncertainty, yet users cannot distinguish between the two. This limitation is exacerbated in multimodal settings, where errors can arise from both visual perception and language reasoning. For example, evaluating a chart-based answer requires visual parsing, data extraction, and reasoning, each introducing compounding uncertainty. Empirically, this reliability gap is significant. On the MLLM-as-a-Judge benchmark (Chen et al., 2024), state-of-the-art VLM judges achieve only 32–34% exact agreement with human ratings on a 5-point scale, and 24–30% of predictions deviate by 2 or more points. Correlation metrics remain modest ($\rho = 0.30\text{--}0.46$), providing limited assurance about per-instance reliability. Existing evaluation metrics thus fail to indicate when a judge’s score can be trusted.

We address this problem using conformal prediction (Vovk et al., 2005), a post-hoc, distribution-free framework that converts point scores into calibrated prediction intervals with provable coverage guarantees. Given a target coverage (e.g., 90%), these intervals contain the true human score with high probability, without requiring model retraining or assumptions on error distributions. Conformal prediction operates directly on score-token log-probabilities, making it readily applicable to existing VLM judge pipelines. While prior work (Sheng et al., 2025) demonstrates the effectiveness of conformal prediction for text-only LLM judges, extending it to multimodal evaluation introduces new challenges:

(i) cross-modal uncertainty from visual and textual reasoning, (ii) discrete single-annotator ground truth, (iii) substantial task heterogeneity across visual domains, and (iv) varying levels of log-probability access across models. Standard metrics such as correlation and accuracy fail to capture evaluation reliability.

We present a systematic study of conformal prediction for VLM-based evaluation. Our contribution is not the application of conformal prediction itself, but the identification of task-dependent uncertainty structure and a fundamental decoupling between ranking quality and scoring reliability in multimodal evaluation. We show that high correlation does not imply reliable absolute scores, and that uncertainty is primarily driven by task ambiguity and annotation quality rather than model scale. Our key findings are:

- **Task-dependent uncertainty:** Prediction interval width varies by up to 70% across tasks. Vision-heavy tasks such as charts and mathematical reasoning yield substantially wider intervals than natural image or aesthetics tasks, providing a quantitative reliability map.
- **Ranking-scoring decoupling:** We identify a failure mode not captured by standard evaluation metrics where judges achieve strong ranking correlation while producing wide, uninformative intervals. That is, they correctly order responses but fail to assign reliable absolute scores.
- **Role of data quality:** Interval width is primarily driven by task difficulty and annotation quality rather than the conformal method. The same judge and method produce $4.5\times$ narrower intervals on a clean, multi-annotator captioning benchmark.
- **Task-conditional calibration:** We show that conditioning on task difficulty (Mondrian CP) yields narrower intervals for easy tasks while improving coverage for hard tasks.

2 Related work

LLM-as-a-Judge. Using LLMs as automated evaluators has become a standard approach for assessing natural language generation (Zheng et al., 2023; Liu et al., 2023). G-Eval (Liu et al., 2023) uses chain-of-thought prompting with GPT-4 to score text quality and improves correlation with human judgments over metrics such as ROUGE and BERTScore. MT-Bench and Chatbot Arena (Zheng et al., 2023) establish pairwise evaluation and reveal systematic biases including position and verbosity bias. SocREval (He et al., 2024) focuses on reasoning.

VLM-as-a-Judge. This paradigm has been extended to multimodal evaluation. Chen et al. (2024) introduce the MLLM-as-a-Judge benchmark with 14 visual task categories and report limited alignment with human judgments. LLaVA-Critic (Xiong et al., 2025) trains a dedicated multimodal judge, while VL-RewardBench (Li et al., 2025) shows that a large fraction of errors are perceptual rather than reasoning-driven.

Uncertainty in LLM-based evaluation. Recent work studies confidence and uncertainty in LLM judgments. Wagner et al. (2024) derive confidence scores from token probabilities and show weak correlation with accuracy. Xie et al. (2024) use token probabilities for calibration and propose boosted conformal methods. Other approaches rely on self-reported confidence (Xiong et al., 2024) or consistency across multiple generations (Tian et al., 2023), both with known limitations. Sheng et al. (2025) apply conformal prediction to text-only LLM judges and show that calibrated intervals improve reliability.

Conformal prediction. Conformal prediction provides distribution-free uncertainty estimates with coverage guarantees (Vovk et al., 2005). Prior work applies it mainly to classification tasks such as multiple-choice QA (Kumar et al., 2023; Su et al., 2024), factual consistency (Quach et al., 2024; Mohri & Hashimoto, 2024), and calibrated risk control (Angelopoulos et al., 2021).

3 Uncertainty Quantification for VLM-based Judge

3.1 VLM-as-a-Judge framework

In the VLM-as-a-Judge setting, a vision-language model M evaluates the quality of a response given an image I and question q . For a candidate answer a , the model produces a

94 score y_0 on a discrete Likert scale $\{1, \dots, K\}$ along with token logits:

$$M(I, q, a) = (z, y_0). \quad (1)$$

95 We extract the log-probabilities of each rating token at the score position, forming a K -
96 dimensional feature vector:

$$\mathbf{x} = [\log P("1"), \log P("2"), \dots, \log P("K")]. \quad (2)$$

97 This feature vector captures the model’s distribution over possible scores. A concentrated
98 distribution, where one score has high probability, indicates high confidence, while a diffuse
99 distribution indicates uncertainty. We also considered additional features derived from
100 entropy, margin of the score distribution, and actor-model entropy. These did not improve
101 performance over $\mathbf{x}$ alone, so we use this representation throughout.

102 3.2 Conformal prediction

103 Conformal prediction (Vovk et al., 2005) is a distribution-free framework for constructing
104 prediction intervals with coverage guarantees. Given a calibration set $\{(\mathbf{x}_i, y_i)\}_{i=1}^n$ and a
105 nonconformity score function $s(\mathbf{x}, y)$, we compute the quantile $\hat{q}$ at miscoverage level α . For
106 a test point $\mathbf{x}$, the prediction interval is

$$\mathcal{C}(\mathbf{x}) = [\hat{y} - \hat{q}, \hat{y} + \hat{q}], \quad (3)$$

107 where $\hat{y} = f(\mathbf{x})$ is a point predictor trained on the calibration set. Under the exchangeability
108 assumption, this interval satisfies

$$\mathbb{P}(y \in \mathcal{C}(\mathbf{x})) \geq 1 - \alpha. \quad (4)$$

109 Conformal prediction is well-suited to this setting because it is post-hoc and does not
110 require retraining or fine-tuning of the judge model, and it is distribution-free, avoiding
111 assumptions on error distributions that may vary across multimodal tasks.

112 3.3 Extensions and diagnostics

113 **Boundary adjustment for discrete scales.** Ground truth scores are discrete integers, while
114 conformal prediction produces continuous intervals. Following Sheng et al. (2025), we apply
115 boundary adjustment:

$$s'(\mathbf{x}, y) = \begin{cases} s(\mathbf{x}, \lceil y \rceil) & \text{if } y \leq \lfloor \hat{y} \rfloor, \\ s(\mathbf{x}, y) & \text{otherwise,} \\ s(\mathbf{x}, \lfloor y \rfloor) & \text{if } y \geq \lceil \hat{y} \rceil, \end{cases} \quad (5)$$

116 which maps the interval $[l, u]$ to $[\lceil l \rceil, \lfloor u \rfloor]$, aligning endpoints with valid rating labels while
117 preserving or increasing coverage.

118 **Task-conditional conformal prediction.** Standard conformal prediction uses a single
119 global quantile. To account for task-dependent variability, Mondrian conformal predic-
120 tion partitions data into groups and computes group-specific quantiles, yielding group-
121 conditional guarantees:

$$\mathbb{P}(y \in \mathcal{C}(\mathbf{x}) \mid G = g) \geq 1 - \alpha. \quad (6)$$

122 This produces tighter intervals for easier tasks and more conservative intervals for harder.
123 We partition the 14 MLLM-Judge datasets into three groups based on per-dataset R2CCP
124 width from an initial standard CP run: *easy* (width < 2.5 : AesBench, MM-Vet, WIT, COCO),
125 *medium* (2.5–3.2: Mind2Web, Conceptual Captions, TextVQA, LLaVA-Bench, VisitBench,
126 ChartQA), and *hard* (> 3.2 : ScienceQA, MathVista, DiffusionDB, InfographicsVQA). Group
127 boundaries are determined once on a held-out pilot split and fixed for all subsequent
128 experiments.

129 **Ranking-Scoring Gap.** Standard metrics capture ranking quality but not the reliability of
 130 absolute scores. We define the Ranking-Scoring Gap:

$$\text{RSG}(d) = |\rho_d| - \left(1 - \frac{w_d}{K-1}\right), \quad (7)$$

131 where ρ_d is the Pearson correlation and w_d is the average interval width. The second
 132 term measures interval informativeness. High RSG indicates strong ranking but unreliable
 133 absolute scores. Following Sheng et al. (2025), we also evaluate the midpoint $\hat{y}_{\text{mid}} =$
 134 $(l + u)/2$ as a calibrated estimate and assess its effectiveness in multimodal settings.

135 4 Experimental setup

136 **Datasets and Judges.** We use two multimodal evaluation benchmarks with human-
 137 annotated ground truth scores.

138 (i) **MLLM-as-a-Judge (Chen et al., 2024).** This benchmark contains 5,717 instances spanning
 139 14 visual task categories. Each instance includes an image, a question, a model-generated
 140 answer, and a human-annotated score on a 1–5 Likert scale from a single annotator. It covers
 141 diverse tasks including visual reasoning, chart understanding, web-based queries, and
 142 aesthetic evaluation. The score distribution is skewed toward higher values, which affects
 143 calibration and conditional coverage.

144 (ii) **Polaris (Wada et al., 2024).** This captioning benchmark contains 8,726 image-caption
 145 pairs with scores aggregated from multiple annotators and mapped to a 1–5 scale. Compared
 146 to MLLM-as-a-Judge, Polaris represents a single, well-defined task with lower annotation
 147 noise, allowing isolation of the effects of task difficulty and label quality on interval width.

148 We evaluate three VLM judges spanning different architectures, scales, and access settings:
 149 (i) **LLaVA-Critic-7B (Xiong et al., 2025)**, an open-source 7B model specialized for multimodal
 150 evaluation; (ii) **Phi-4-reasoning-vision-15B (Abdin et al., 2025)**, a 15B reasoning model
 151 with extended chain-of-thought generation; and (iii) **Gemini 2.5 Flash (Google DeepMind,**
 152 **2025)**, a closed-source API model with strong ranking performance but highly confident
 153 predictions. All models use a consistent evaluation prompt across datasets (Appendix A.3),
 154 enabling controlled comparison across architectures.

155 **Evaluation protocol.** Following Sheng et al. (2025), we randomly split each dataset 50/50
 156 into calibration and test sets across 10 random seeds and report mean $\pm$ standard deviation.
 157 We use significance level $\alpha = 0.10$, targeting 90% coverage. This yields approximately 2,859
 158 calibration and 2,858 test samples per split for MLLM, and 4,363/4,363 for Polaris. For each
 159 instance, we extract score-token log-probabilities to form a 5-dimensional feature vector.
 160 Additional implementation details are provided in Appendix A.2.

161 We report the following metrics. Coverage measures the fraction of test samples where
 162 the ground truth lies within the prediction interval, with a target of ≥ 0.90 . Width is the
 163 average interval size on the 1–5 scale (range 0 to 4), where lower is better given sufficient
 164 coverage. Point prediction quality is evaluated using Pearson ρ , Spearman ρ_s , and Kendall
 165 τ correlations, along with exact-match accuracy, relaxed ± 1 accuracy, and mean absolute
 166 error. We also report judge bias as the mean signed error (judge score minus human
 167 ground truth), stratified by ground truth level. Both raw (continuous) and boundary-
 168 adjusted (discrete) intervals are reported for all methods. For per-dataset analysis, we run
 169 R2CCP independently on each of the 14 datasets to evaluate conditional coverage and
 170 task-dependent interval widths.

171 5 Results and Discussions

172 Before examining judge uncertainty, we establish baseline point prediction performance for
 173 all three judges on MLLM-as-a-Judge (Table 1). All three judges achieve comparable exact
 174 accuracy (32–34%) but differ substantially in other metrics. Gemini achieves the highest
 175 correlation (Pearson = 0.459), Phi-4 the lowest MAE (1.005) and lowest bias (+0.08), and

Table 1: Cross-judge comparison on MLLM-as-a-Judge (10 seeds). Metrics include point prediction, score calibration, and conformal prediction results.

Group	Metric	LLaVA-Critic 7B	Phi-4 15B	Gemini 2.5 Flash
Point pred.	Pearson ρ	.402	.303	.459
	Spearman ρ_s	.356	.293	.446
	Kendall τ	.300	.251	.377
	Accuracy	32.2%	34.2%	32.1%
	± 1 Acc.	75.1%	76.1%	70.3%
	MAE	1.031	1.005	1.122
	Bias	+0.38	+0.08	-0.05
Score cal.	Overconf. (>0.99)	25.6%	43.0%	83.0%
	Overconf. (>0.999)	6.6%	16.9%	77.4%
R2CCP	Coverage (raw)	.900	.891	.898
	Width (raw)	3.05	3.13	2.85
	Coverage (adj.)	.981	.981	.980
	Width (adj.)	3.60	3.70	3.41
CHR	Coverage (raw)	.880	.896	.880
	Width (raw)	2.97	3.16	2.80

LLaVA-Critic produces the most calibrated log-probability distributions (25.6% overconfident). These differences directly affect the quality of VLM-judge uncertainty prediction, shaping both interval width and coverage.

5.1 Conformal methods comparison

Table 2 reports coverage and interval width for all 8 conformal prediction methods on MLLM-as-a-Judge using LLaVA-Critic-7B. The methods fall into three performance tiers.

R2CCP achieves coverage closest to the 90% target (90.0%) with width 3.05, providing the best balance between calibration and interval size. CHR produces slightly narrower intervals (2.97) but under-covers (88.0%), while Boosted LCP yields stable intervals but also under-covers. After boundary adjustment, all three exceed 96% coverage. In contrast, CQR, Asymmetric CQR, and OrdinalAPS collapse to near-full-range intervals (~ 4.0), achieving near-100% coverage but providing little discrimination. After boundary adjustment, CQR variants produce constant intervals $[1, 5]$ for all samples.

This difference arises from how methods use the input features. Approaches based on conditional quantile estimation degrade when features are weak predictors, leading to uniformly wide intervals. In contrast, R2CCP models the full conditional distribution $P(y | x)$, which is more robust in this setting. Boundary adjustment is disproportionately important in our setting: it increases coverage by 8+ percentage points on our 5-class integer GT, compared to 2–5 points reported by Sheng et al. (2025) on 13-class averaged GT, because continuous interval endpoints frequently fall between our sparse integer labels. Based on these results, we use **R2CCP** as the default method due to its accurate coverage and reasonable interval width. CHR is preferred when narrower intervals are desired and slight under-coverage is acceptable.

5.2 Cross-judge comparison

Table 1 compares the three VLM judges on both point prediction and conformal prediction metrics, revealing several non-obvious patterns.

Despite producing highly overconfident score distributions (83% of softmax maxima > 0.99), Gemini yields the narrowest conformal intervals (2.85 raw, 3.41 adjusted), outperforming both LLaVA-Critic and Phi-4. This apparent paradox highlights a key property of conformal prediction: interval quality depends on the discriminative signal in the feature representation rather than the calibration of the softmax distribution. Even when probabilities are sharply peaked, relative differences between score-token log-probabilities remain infor-

Table 2: Conformal prediction methods on MLLM-as-a-Judge (LLaVA-Critic-7B, $\alpha = 0.10$, 10 seeds). Methods grouped by performance tier. Bold indicates best within well-calibrated.

Method	Cov. (raw)	Width (raw)	Cov. (adj.)	Width (adj.)
<i>Tier 1: Well-calibrated (coverage $\approx 90\%$, informative width)</i>				
R2CCP	.900$\pm$.016	3.05 $\pm$.10	.981 $\pm$.005	3.60 $\pm$.09
CHR	.880 $\pm$.010	2.97$\pm$.07	.965 $\pm$.006	3.43$\pm$.10
Boosted LCP	.863 $\pm$.012	3.02 $\pm$.03	.977 $\pm$.004	3.51 $\pm$.04
<i>Tier 2: Moderate (slight over/under-coverage, moderate width)</i>				
Naive Split CP	.895 $\pm$.009	3.23 $\pm$.06	.990 $\pm$.003	3.78 $\pm$.04
LVD	.894 $\pm$.009	3.21 $\pm$.11	.988 $\pm$.005	3.71 $\pm$.11
Boosted CQR	.878 $\pm$.009	3.54 $\pm$.09	.997 $\pm$.002	3.93 $\pm$.04
<i>Tier 3: Degenerate (massive over-coverage, uninformative)</i>				
CQR	.996 $\pm$.005	3.95 $\pm$.06	1.00 $\pm$.000	4.00 $\pm$.00
Asym. CQR	.996 $\pm$.005	3.95 $\pm$.06	1.00 $\pm$.000	4.00 $\pm$.00
OrdinalAPS	.999 $\pm$.000	3.99 $\pm$.00	.999 $\pm$.000	3.99 $\pm$.00

Table 3: Per-dataset R2CCP analysis for LLaVA-Critic-7B on MLLM-as-a-Judge (10 seeds). Datasets sorted by raw width. Width varies 70% across task types—from 2.08 (AesBench) to 3.50 (InfographicsVQA).

Dataset	N	Width (raw)	Width (adj.)	Pearson	± 1 Acc.	Category
AesBench	392	2.08	2.66	.402	88.3%	Aesth.
MM-Vet	258	2.18	2.76	.260	80.2%	Gen. VQA
WIT	399	2.38	3.15	.164	78.7%	Know.
COCO	397	2.43	3.07	.362	84.9%	Gen. VQA
Mind2Web	398	2.69	3.31	.268	74.4%	Know.
Conc. Cap.	398	2.70	3.38	.362	81.7%	Know.
TextVQA	399	2.81	3.28	.389	79.2%	Vision
LLaVA-B.	396	2.92	3.61	.244	81.1%	Gen. VQA
VisitBench	397	2.96	3.48	.352	75.1%	Gen. VQA
ChartQA	400	3.08	3.57	.507	73.2%	Vision
ScienceQA	396	3.27	3.64	.258	69.7%	Vision
MathVista	790	3.37	3.71	.376	65.1%	Vision
DiffusionDB	299	3.41	3.89	.089	57.2%	Aesth.
Infograph.	398	3.50	3.84	.411	69.3%	Vision

208 mative, enabling R2CCP to construct tight and valid intervals. Overconfidence in raw
209 probabilities therefore does not imply poor uncertainty estimation.

210 The three judges also exhibit a clear tradeoff between accuracy and correlation. Phi-4
211 achieves the highest exact accuracy (34.2%) and lowest MAE (1.005) but the lowest Pearson
212 correlation (0.303), while Gemini achieves the highest correlation (0.459) but lower ± 1
213 accuracy (70.3%) and higher MAE (1.122). This divergence reflects two distinct aspects of
214 judge quality: correlation captures ranking ability, while accuracy reflects absolute scoring
215 precision. Gemini is a stronger ranker, whereas Phi-4 is more precise in absolute scoring.

216 This finding has a concrete mechanism: R2CCP learns a neural network mapping from
217 the 5-dimensional logprob vector to nonconformity scores. Even when Gemini’s softmax
218 distribution is sharply peaked, the *relative* magnitudes across the five score tokens still carry
219 discriminative information—for instance, whether the second-highest logprob is for an
220 adjacent score or a distant one. R2CCP exploits these relative patterns regardless of absolute
221 calibration. Overall, conformal interval quality is governed by the signal-to-noise ratio of
222 the representation rather than probability calibration.

Table 4: Judge bias by ground truth score level. All judges overscore bad answers (GT=1,2) and underscore excellent ones (GT=5). Gemini has the lowest bias for bad answers.

GT Score	LLaVA-Critic	Phi-4	Gemini
GT=1 (N=641)	+1.73	+1.98	+ 0.89
GT=2 (N=707)	+1.21	+1.13	+ 0.43
GT=3 (N=1263)	+0.79	+0.34	+ 0.23
GT=4 (N=1791)	+0.10	-0.35	- 0.15
GT=5 (N=1314)	-0.74	-1.04	- 0.87
Overall	+0.38	+0.08	- 0.05

5.3 Error structure and bias

Table 4 shows that VLM judges exhibit structured and systematic errors. Low-quality responses (GT=1) are consistently overscored, often assigned scores near 3, while high-quality responses are underscored, leading to compression toward the middle of the scale. This pattern is consistent across all three judges. These structured errors explain the observed uncertainty behavior. Large and inconsistent deviations increase nonconformity scores, leading to wider intervals, particularly for vision-heavy tasks. In contrast, tasks with more consistent predictions, even if biased, produce narrower intervals. This shows that interval width reflects not only accuracy but the distribution and magnitude of errors.

5.4 Task-dependent uncertainty

Table 3 reveals strong task dependence in judge uncertainty, with prediction interval width varying substantially across categories. AesBench (aesthetics assessment) yields the narrowest intervals (2.08), while InfographicsVQA produces the widest (3.50), a 68% relative difference on a 4-point scale. Equivalently, intervals cover 52% of the score range for AesBench versus 88% for InfographicsVQA. This variation provides a quantitative reliability map, showing that the informativeness of VLM-based evaluation depends strongly on task.

Aggregating by meta-category (Table 6) reveals a clear structure. Vision-heavy tasks produce substantially wider intervals than Knowledge/Web tasks (3.21 vs. 2.59), reflecting the difficulty of structured visual reasoning such as charts, diagrams, and infographics. Notably, these tasks also exhibit higher correlation, indicating that judges can rank responses but struggle to assign precise absolute scores, a pattern we analyze through ranking-scoring decoupling in §5.5. This pattern is consistent across all three judges (Figure 1). AesBench consistently yields the narrowest intervals and InfographicsVQA the widest. The Spearman rank correlation of per-dataset widths is 0.93 between LLaVA-Critic and Phi-4, and 0.82 between LLaVA-Critic and Gemini, indicating that interval width is primarily determined by the task rather than the specific model.

Overall, interval width reflects the combined effect of task difficulty, score distribution, and visual diversity. Tasks with higher accuracy and more consistent structure yield narrower intervals, while those requiring complex visual reasoning produce broader uncertainty. The observed task-dependent variability can be exploited by group-conditional conformal methods. Using Mondrian conformal prediction with task difficulty groups, we obtain narrower intervals for easy tasks (16.6% reduction) while improving coverage on hard tasks, with similar overall coverage. This confirms that task-dependent uncertainty is structured and can be leveraged for more efficient calibration (Table 5).

5.5 Ranking-scoring decoupling

Table 3 compares per-dataset correlation with interval width. ChartQA achieves the highest Pearson correlation among all 14 datasets ($\rho = 0.507$) while producing among the widest intervals (3.08). In contrast, WIT has low correlation ($\rho = 0.164$) but relatively narrow intervals (2.38). This ranking-scoring decoupling exposes a failure mode that standard evaluation metrics do not capture. A judge can correctly order responses while assigning

Table 5: Mondrian CP vs. standard R2CCP (LLaVA-Critic-7B, boundary-adjusted, 10 seeds). Mondrian narrows easy-task intervals by 16.6% while improving hard-task coverage.

Group	Standard R2CCP		Mondrian R2CCP	
	Cov.	Width	Cov.	Width
Easy	99.0%	3.55	97.5%	2.96 (−16.6%)
Medium	99.1%	3.60	98.7%	3.49 (−3.1%)
Hard	96.7%	3.63	98.2%	3.78 (+4.1%)
Overall	98.1%	3.60	98.2%	3.47 (−3.6%)

Table 6: Average R2CCP interval width by task meta-category (LLaVA-Critic-7B, raw intervals). Vision-Heavy tasks produce 24% wider intervals than Knowledge/Web tasks.

Category	N	Avg. Width	Avg. Pearson	Avg. ± 1 Acc.
Knowledge/Web	1,195	2.59	.265	78.3%
General VQA	1,448	2.62	.304	80.3%
Aesthetics/AI	691	2.75	.246	72.8%
Vision-Heavy	2,383	3.21	.388	71.3%

unreliable absolute scores. For ChartQA, high correlation suggests strong performance, yet wide intervals indicate poor calibration of absolute scores. Notably, this failure mode is not well captured in standard evaluation metrics.

The effect arises because ranking depends only on monotonic consistency, while precise scoring requires accurate calibration of score magnitudes. On tasks such as chart understanding, the judge can distinguish relative quality but struggles to assign consistent absolute scores. This distinction has direct implications. For tasks where relative ordering is sufficient, correlation-based evaluation remains appropriate. For applications requiring reliable absolute scores, wide intervals indicate that point predictions are unreliable, and pairwise comparison is preferable. The decoupling is task-specific and cannot be inferred from aggregate metrics. The same pattern appears across models. For example, Gemini achieves high correlation with tight intervals on TextVQA ($\rho = 0.600$, width = 2.33), but low correlation with similarly wide intervals on Mind2Web ($\rho = 0.065$, width = 2.92), confirming that ranking quality and scoring precision are not inherently coupled.

5.6 Effect of ground truth quality: MLLM-Judge vs. Polaris

Table 7 compares interval widths under identical conditions, using the same judge (LLaVA-Critic-7B) and conformal method (R2CCP with $\alpha = 0.10$). The results differ substantially: width is 3.05 on MLLM-Judge (61% of the score range) and 0.68 on Polaris (14%), a $4.5\times$ reduction, while both achieve the target 90% coverage. This difference shows that interval width is primarily determined by the evaluation task and ground truth quality rather than the conformal method or model. Polaris is a single, well-defined captioning task with multi-annotator averaged labels, whereas MLLM-Judge spans diverse visual reasoning tasks with single-annotator scores. The resulting difference in signal quality is reflected directly in conformal intervals. The judge achieves substantially higher accuracy on Polaris (Pearson = 0.906 vs. 0.402), leading to concentrated nonconformity scores and tighter intervals.

Implications. When the task is well-defined and annotations are reliable, conformal prediction produces tight and actionable uncertainty estimates. Conversely, wide intervals on MLLM-Judge reflect genuine task difficulty and label noise rather than a limitation of the method. Notably, this advantage is judge-dependent: Phi-4-15B achieves only Pearson = 0.128 on Polaris despite being twice the size of LLaVA-Critic, demonstrating that model specialization for evaluation matters more than scale (Appendix F.4). Interval width should be interpreted as an indicator of evaluation reliability.

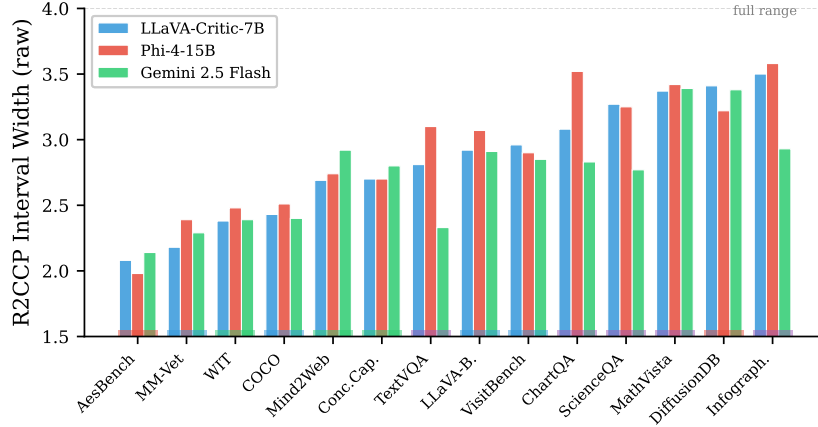


Figure 1: R2CCP interval width across 14 task categories for all three judges. Task-dependent uncertainty is consistent: aesthetics tasks yield tight intervals while vision-heavy tasks produce wide intervals regardless of the judge. The ordering is largely preserved across judges (Spearman $\rho = 0.82\text{--}0.93$).

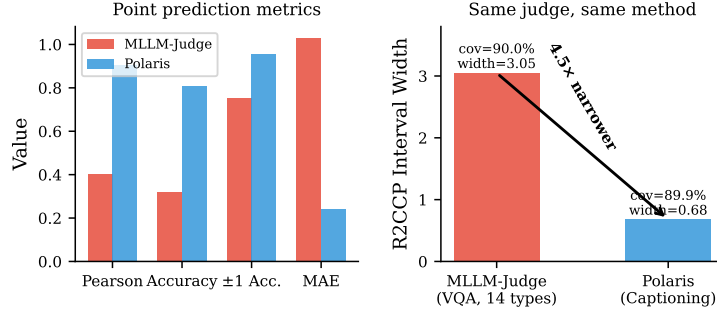


Figure 2: MLLM-Judge vs. Polaris: same judge (LLaVA-Critic-7B), same CP method (R2CCP, $\alpha = 0.10$), but $4.5\times$ narrower intervals when ground truth is clean (multi-annotator) and the task is well-defined (captioning).

Multi-judge feature fusion. We also test whether combining logprob features from multiple judges improves conformal prediction. Contrary to expectation, fusion degrades performance: combining all three judges produces wider intervals (3.14) than using the best single judge (Gemini, 2.85). This occurs because additional features from weaker judges introduce noise without sufficient signal, making the learned nonconformity mapping less effective under limited calibration data. These results suggest that feature quality is more important than feature quantity, i.e., using a single strong judge is preferable to multiple weaker ones.

6 Conclusion

We show that VLM judge reliability is governed primarily by task structure and data quality rather than the model or method. Uncertainty varies strongly across tasks and is consistent across judges. We identify a ranking-scoring decoupling where judges can rank responses correctly while assigning unreliable absolute scores, a failure mode not captured by standard metrics, and show that interval width is driven by task difficulty and annotation quality, with a $4.5\times$ reduction under clean, well-defined conditions. These findings lead to a simple operational guideline: narrow conformal intervals indicate reliable evaluation suitable for absolute scoring, while wide intervals signal ambiguity or noise where relative ranking or pairwise comparison is more appropriate. Conformal prediction thus provides a practical, uncertainty-aware framework for deploying VLM judges, with interval width serving as a direct indicator of evaluation reliability.

Table 7: Same judge (LLaVA-Critic-7B) and method (R2CCP, $\alpha = 0.10$). Multi-annotator GT and a well-defined task yield $4.5\times$ narrower intervals, while achieving 90% coverage.

Group	Metric	MLLM-Judge (14 tasks)	Polaris (Captioning)
Data	Samples	5,717	8,726
	GT type	1 annotator, integer	3+ annotators, avg.
	Task types	14 visual tasks	1 task
Point	Pearson ρ	.402	.906
	Accuracy	32.2%	80.9%
	± 1 Accuracy	75.1%	95.4%
	MAE	1.031	0.243
	Bias	+0.38	−0.06
R2CCP	Coverage	.900	.899
	Width	3.05 (61%)	0.68 (14%)

Table 8: Multi-judge feature fusion (R2CCP, 10 seeds). Combining judges does NOT outperform the best single judge (Gemini). Logprob quality > quantity.

Configuration	Feat.	Cov. (raw)	Width (raw)	Width (adj.)
Gemini only	5	.898	2.85	3.41
LLaVA + Gemini	10	.901	3.02	3.51
LLaVA only	5	.900	3.05	3.60
Multi-Judge (all 3)	15	.905	3.14	3.56
LLaVA + Phi-4	10	.903	3.18	3.62
Phi-4 only	5	.891	3.13	3.70

Reproducibility statement

All experiments use publicly available models (LLaVA-Critic-7B, Phi-4-reasoning-vision-15B) and datasets (MLLM-as-a-Judge, Polaris). Gemini 2.5 Flash is accessed via the Google Vertex AI API. We will release our complete code, including inference scripts, feature extraction pipeline, conformal prediction runners, and all result CSVs upon acceptance. Results are reported across 10 random seeds with mean $\pm$ standard deviation. All conformal prediction methods use publicly available implementations (R2CCP from the authors’ wheel, MAPIE v0.8.6 for CQR variants, custom implementations following published algorithms for CHR, LVD, Boosted methods, and OrdinalAPS). Hardware: $2\times$ RTX 6000 Ada GPUs (48GB each) for open-source model inference.

Limitations

On the MLLM-Judge benchmark, 68.8% of boundary-adjusted intervals span more than 75% of the rating scale, providing limited discrimination for individual evaluations. This reflects the genuine difficulty of the task and noisy single-annotator ground truth rather than a method limitation—the same approach yields decisive intervals on Polaris (width 0.68). Conditional coverage for the lowest-quality responses (GT=1) drops to 88.9%, below the 90% target, due to systematic judge overscoring of poor answers. Our score-token extraction heuristic (backward scan for the final digit) was manually verified on 100 samples per judge with 100% accuracy, but may be less reliable on other models with different output formats. Results use a single $\alpha = 0.10$; behavior at stricter significance levels (e.g., $\alpha = 0.05$) remains unexplored. Conformal prediction requires a human-annotated calibration set, which partially offsets the cost savings of automated evaluation; however, Appendix F.2 shows coverage is robust down to 30% calibration splits ($\sim 1,700$ samples). Finally, our evaluation is specific to MLLM-Judge’s 14 visual task categories and the Polaris captioning benchmark; generalization to other domains (e.g., medical imaging, document analysis) requires further validation.

Ethics statement

Our work analyzes the reliability of automated VLM evaluators and does not introduce new evaluation capabilities—it quantifies the uncertainty of existing ones. We note several ethical considerations: (1) Conformal prediction guarantees are marginal (averaged over the calibration/test split) rather than conditional, and should not be over-interpreted as individual-instance confidence. (2) Biased human annotations in the calibration set propagate to biased prediction intervals—conformal prediction cannot correct for systematic annotator bias. (3) The positive bias we observe (judges overscoring bad answers) may mask quality problems in deployed AI systems if interval information is ignored. We encourage practitioners to use conformal intervals as a complement to, not a replacement for, careful human evaluation in high-stakes settings.

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429 A Methods and Implementation

430 A.1 Conformal prediction methods

431 We evaluate eight conformal prediction methods spanning regression-based and
432 classification-based approaches, following [Sheng et al. \(2025\)](#).

Regression-based methods. These include Naive Split Conformal Prediction, which uses absolute residuals; Conformalized Quantile Regression (CQR) (Romano et al., 2019), which learns conditional quantiles; Asymmetric CQR (Sesia & Candès, 2020), which calibrates upper and lower bounds separately; Conformal Histogram Regression (CHR) (Sesia & Romano, 2021), which uses histogram-based estimation; and Locally adaptive variance-based methods (LVD) (Lin et al., 2021). We also evaluate boosted variants of CQR and LCP (Xie et al., 2024), as well as R2CCP (Guha et al., 2024), which models the full conditional distribution via regression-as-classification.

Classification-based methods. We additionally consider Ordinal Adaptive Prediction Sets (OrdinalAPS) (Lu et al., 2022), which construct contiguous prediction sets that respect the ordinal structure of ratings.

A.2 Feature extraction

For each evaluation instance, we extract the log-probabilities of the score tokens as defined in Equation 2. Because chain-of-thought outputs may contain multiple intermediate numeric tokens, identifying the final rating token is non-trivial.

We use the following heuristic:

1. Search for score-related keywords (e.g., “score”, “rating”) followed by a digit token.
2. Perform a backward scan from the end of the generated sequence to locate the final rating token.

At the identified position, we extract the log-probabilities for all rating tokens (“1” through “5”), yielding a 5-dimensional feature vector per sample. For models with partial log-probability access, missing tokens are assigned a default low value.

A.3 Evaluation prompts

A.3.1 MLLM-as-a-Judge CoT prompt

We use a chain-of-thought prompt adapted from the MLLM-as-a-Judge dataset (Chen et al., 2024). The prompt follows the “COT figure” setting that forces the judge to analyze the image first, then evaluate the response, then provide a final score:

Please serve as an unbiased judge in assessing the quality of the response from an AI assistant regarding the user’s instruction and the provided image.

Evaluation Steps: Please examine the provided image attentively. Begin by conducting a comprehensive analysis of the figure provided. Then, utilize the insights from your analysis to critically evaluate the response. Finally, based on your figure analysis and response evaluation, form a well-reasoned judgement.

Scoring Rubric: Poor (1): The response significantly deviates from the user’s instruction and fails to address the query effectively. It shows a lack of relevance, accuracy, and comprehensiveness. Creativity and granularity are absent or poorly executed. Fair (2): The response addresses the user’s instruction partially, with evident shortcomings in relevance, accuracy, or comprehensiveness. It lacks depth in creativity and granularity. Average (3): The response adequately addresses the user’s instruction, showing a fair level of relevance, accuracy, and comprehensiveness. It reflects a basic level of creativity and granularity but may lack sophistication. Good (4): The response is well-aligned with the user’s instruction, demonstrating a high degree of relevance, accuracy, and comprehensiveness. It shows creativity and nuanced understanding. Excellent (5): The response perfectly adheres to the user’s instruction, excelling in relevance, accuracy, comprehensiveness, creativity, and granularity.

Notice: In your evaluation, weigh factors such as relevance, accuracy, comprehensiveness, creativity, and the granularity of the response. Do not allow the length of the response to influence your evaluation. Be as objective as possible.

*Here is the input: [The Start of User Instruction] {question} [The End of User Instruction]
[The Start of Assistant’s Answer] {actor_answer} [The End of Assistant’s Answer]*

Table 9: Top CP methods across judges (raw coverage and width, 10 seeds).

Method	LLaVA-Critic		Phi-4		Gemini	
	Cov.	Width	Cov.	Width	Cov.	Width
R2CCP	.900	3.05	.891	3.13	.898	2.85
CHR	.880	2.97	.896	3.16	.880	2.80
B. LCP	.863	3.02	.875	3.24	.863	2.88
Naive CP	.895	3.23	.897	3.30	.893	3.08
CQR	.996	3.95	.997	3.98	.996	3.95

First, provide your analysis of the image and the response. Then, at the very end of your response, provide your final score in exactly this format on its own line: "Score: X" where X is a single number from 1 to 5.

A.3.2 Polaris captioning prompt

For the Polaris captioning dataset, we adapt the prompt structure for caption evaluation:

Please serve as an unbiased judge in assessing the quality of an image caption generated by an AI model.

Evaluation Steps: Please examine the provided image attentively. Then evaluate how well the generated caption describes the image content.

Scoring Rubric: Poor (1): The caption is completely inaccurate, irrelevant, or fails to describe the image content. Major factual errors or hallucinations are present. Fair (2): The caption partially describes the image but has significant inaccuracies, missing key elements, or includes incorrect details. Average (3): The caption adequately describes the main content of the image with fair accuracy, but may miss some details or include minor inaccuracies. Good (4): The caption accurately describes the image content with good detail and relevance, with only minor omissions. Excellent (5): The caption perfectly describes the image content with high accuracy, comprehensive detail, and excellent relevance to what is shown.

Provide your analysis of how well the caption matches the image, then provide your final score in exactly this format on its own line: "Score: X" where X is a single number from 1 to 5.

B Core Results

B.1 Cross-Judge Comparison

Table 9 compares the top conformal methods across all three judges. R2CCP and CHR consistently achieve the best coverage-width tradeoff. Gemini produces the narrowest intervals across methods, indicating better-calibrated score distributions.

B.2 Per-Dataset Results

Width comparison across judges. Table 10 shows per-dataset interval widths. Gemini yields the narrowest intervals on 10 of 14 datasets, with the largest gains on vision-heavy tasks, indicating better alignment between ranking and scoring.

Relaxed accuracy and MAE. Table 11 reports relaxed accuracy (± 1) and MAE. While higher accuracy generally correlates with narrower intervals, several datasets violate this trend, confirming that ranking quality and scoring precision are not tightly coupled.

Confusion matrices. Table 12 reveals distinct error patterns across judges. LLaVA exhibits broad dispersion, Phi-4 gravitates toward score 4, and Gemini polarizes between scores 1 and 5.

Table 10: Per-dataset R2CCP raw width for all three judges. Bold indicates the narrowest width for each dataset.

Dataset	LLaVA	Phi-4	Gemini	Cat.
AesBench	2.08	1.98	2.14	Aesth.
MM-Vet	2.18	2.39	2.29	Gen.
WIT	2.38	2.48	2.39	Know.
COCO	2.43	2.51	2.40	Gen.
Mind2Web	2.69	2.74	2.92	Know.
Conc. Cap.	2.70	2.70	2.80	Know.
TextVQA	2.81	3.10	2.33	Vision
LLaVA-B.	2.92	3.07	2.91	Gen.
VisitBench	2.96	2.90	2.85	Gen.
ChartQA	3.08	3.52	2.83	Vision
ScienceQA	3.27	3.25	2.77	Vision
MathVista	3.37	3.42	3.39	Vision
DiffusionDB	3.41	3.22	3.38	Aesth.
Infograph.	3.50	3.58	2.93	Vision
Overall	3.05	3.13	2.85	—

Table 11: Per-dataset relaxed accuracy and MAE for LLaVA-Critic-7B.

Dataset	N	Exact	± 1	MAE	Bias
AesBench	392	41.3%	88.3%	0.732	+0.49
COCO	397	34.5%	84.9%	0.854	+0.04
Conc. Cap.	398	35.4%	81.7%	0.872	+0.60
LLaVA-B.	396	31.8%	81.1%	0.922	+0.55
MM-Vet	258	37.2%	80.2%	0.926	−0.01
TextVQA	399	36.3%	79.2%	0.942	+0.30
WIT	399	33.8%	78.7%	0.912	+0.51
VisitBench	397	31.7%	75.1%	1.018	+0.44
Mind2Web	398	29.6%	74.4%	1.008	+0.19
ChartQA	400	30.2%	73.2%	1.085	+0.31
ScienceQA	396	26.5%	69.7%	1.200	+0.30
Infograph.	398	25.1%	69.3%	1.191	+0.18
MathVista	789	32.3%	65.1%	1.235	+0.33
DiffusionDB	299	23.7%	57.2%	1.385	+1.23

518 B.3 Polaris Results

519 **Confusion matrix.** Table 13 shows near-perfect identification of low-quality captions (GT =
520 1), with most confusion between adjacent categories.

521 **Error-bin CP coverage.** Table 14 shows that tight intervals reduce raw coverage for ± 1
522 errors, but boundary adjustment restores coverage to near-perfect levels.

523 C Diagnostics and Analysis

524 **Error Analysis.** Figure 3 and Table 15 show conformal prediction behavior conditioned on
525 judge error magnitude. CP is most effective when the judge is wrong: boundary-adjusted
526 coverage exceeds 99% for ± 1 and ± 2 errors, which constitute the majority of mistakes.
527 Overall, CP recovers 97.8% of all judge errors.

528 **Bias Analysis.** Table 16 shows mean signed error conditioned on ground truth score. All
529 judges overscore low-quality responses and underscore high-quality responses, produc-
530 ing compression toward mid-range scores. This bias explains the asymmetry in error
531 distributions.

Table 12: Row-normalized confusion matrices (% , GT rows, Pred columns).

	P=1	P=2	P=3	P=4	P=5	± 1
<i>LLaVA-Critic-7B</i>						
GT=1	32.6	12.9	17.0	23.4	14.0	45.6
GT=2	17.0	11.7	21.9	32.2	17.1	50.6
GT=3	5.2	7.0	18.6	42.0	27.2	67.6
GT=4	3.3	3.6	13.4	38.9	40.8	93.1
GT=5	1.9	2.2	10.7	38.4	46.7	85.2

Table 13: Confusion matrix for Polaris (%).

	P=1	P=2	P=3	P=4	P=5	± 1
GT=1	99.6	0.1	0.2	0.1	0.0	99.7
GT=2	74.2	12.9	5.7	4.8	2.4	92.7
GT=3	18.6	13.4	32.1	30.8	5.2	76.2
GT=4	1.6	4.0	14.6	64.2	15.6	94.4
GT=5	0.8	2.0	5.6	63.9	27.8	91.7

532 **Ranking-Scoring Decoupling.** Table 17 and Figure 4 show that ranking quality and scoring
533 precision are not coupled. High correlation does not imply narrow intervals.

534 **Conditional Coverage.** Table 18 shows coverage conditioned on ground truth score. Cover-
535 age is lowest for GT = 1 due to systematic overscoring, while other levels achieve near-perfect
536 adjusted coverage.

537 D Ablations

538 **Mondrian Conformal Prediction.** Table 5 in the main paper shows results for Mondrian
539 conformal prediction, which computes group-specific quantiles based on task difficulty.
540 Compared to standard R2CCP, Mondrian CP produces narrower intervals for easier tasks
541 and improved coverage for harder ones. Easy tasks receive 16.6% narrower intervals, while
542 hard tasks achieve higher coverage, reducing unnecessary conservativeness.

543 **Naive Split CP vs. R2CCP.** Table 19 shows that R2CCP consistently outperforms the Naive
544 Split CP baseline across all datasets, producing uniformly narrower intervals.

545 **Effect of Chain-of-Thought Prompting.** Table 20 shows that CoT improves correlation
546 metrics and significantly reduces overconfidence, providing better signal for conformal
547 prediction.

548 E Additional Observations

549 **Interval Informativeness.** On MLLM-Judge, 0.5% of intervals are decisive, 30.7% are
550 moderately informative, and 68.8% are uninformative. In contrast, Polaris produces mostly
551 decisive intervals due to stronger signal quality.

552 **Midpoint Analysis.** Table 21 shows that midpoint-based estimates do not consistently
553 outperform raw scores due to wide intervals.

554 F Robustness and Sensitivity

555 F.1 Seed sensitivity

556 Table 22 shows that R2CCP results stabilize by 5–10 seeds. Coverage and width means
557 remain virtually unchanged from 5 to 30 seeds, with standard deviation decreasing modestly.
558 Our choice of 10 seeds provides a reliable estimate with manageable compute.

Table 14: CP coverage by error magnitude on Polaris.

Error	% samples	Cov. (raw)	Cov. (adj.)
Exact (0)	80.9%	98.2%	99.6%
± 1	14.6%	58.1%	99.4%
± 2	4.1%	50.0%	80.7%
± 3	0.5%	0.6%	52.4%

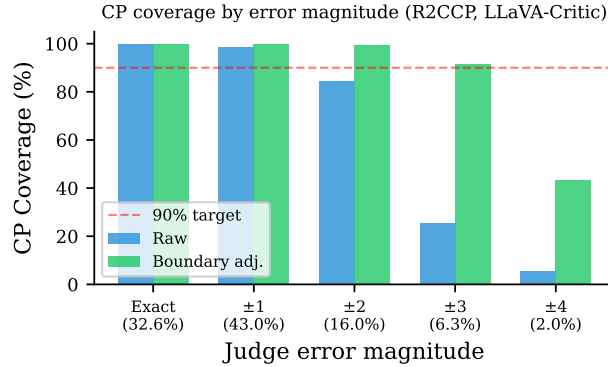


Figure 3: CP coverage by error magnitude. Boundary adjustment is critical: for ± 2 errors, raw coverage is 84% but adjusted coverage reaches 99%.

559 F.2 Calibration set size

560 Table 23 shows that R2CCP is robust to calibration/test split ratio. Coverage remains near
 561 the 90% target across all splits, with width stable at ~ 3.05 . With a 30/70 split (smallest
 562 calibration set), coverage drops slightly to 88.7% but remains close to target. The 50/50 split
 563 used throughout our experiments provides a balanced trade-off.

564 F.3 Cross-judge error-bin coverage

565 Table 24 extends the error-bin analysis to all three judges. The recovery pattern is consistent:
 566 boundary-adjusted coverage exceeds 99% for ± 1 and ± 2 errors across all judges. Gemini
 567 achieves higher coverage for larger errors (± 3 : 96.2% vs. 91.5% for LLaVA) because its
 568 narrower but better-centered intervals extend to the true score more often.

569 F.4 Phi-4 on Polaris: model specialization vs. scale

570 Table 25 compares judges on the Polaris captioning benchmark. LLaVA-Critic-7B, trained
 571 specifically for evaluation, dramatically outperforms the general-purpose Phi-4-15B. With
 572 chain-of-thought reasoning, Phi-4 achieves only Pearson = 0.128 and R2CCP width = 3.60.
 573 Without reasoning mode, performance degrades further to Pearson = 0.036, as the model
 574 defaults to score 4 for 72% of samples. This confirms that **evaluation specialization matters**
 575 **more than model scale**: a 7B specialist produces $5.3\times$ narrower conformal intervals than a
 576 15B generalist on the same task.

Table 15: CP coverage by error magnitude (R2CCP, LLaVA-Critic-7B).

Error	% samp.	Cov. (raw)	Cov. (adj.)	Width (raw)	Width (adj.)
0 (exact)	32.6%	99.8%	100%	2.98	3.55
± 1	43.0%	98.7%	99.9%	3.08	3.63
± 2	16.0%	84.3%	99.4%	3.12	3.65
± 3	6.3%	25.4%	91.5%	3.05	3.61
± 4	2.0%	5.7%	43.3%	2.89	3.42
<i>Overall: 97.8% recovery</i>					

Table 16: Mean signed error by ground truth score level.

GT Score	LLaVA	Phi-4	Gemini
1	+1.73	+1.98	+0.89
2	+1.21	+1.13	+0.43
3	+0.79	+0.34	+0.23
4	+0.10	−0.35	−0.15
5	−0.74	−1.04	−0.87
Overall	+0.38	+0.08	−0.05

Table 17: Ranking-Scoring Gap (RSG). Positive values indicate strong ranking but weak scoring precision.

Dataset	LLaVA		Phi-4		Gemini	
	ρ	RSG	ρ	RSG	ρ	RSG
Infograph.	.411	+.287	.162	+.058	.547	+.281
ChartQA	.507	+.276	.261	+.141	.494	+.201
MathVista	.376	+.219	.339	+.194	.414	+.262
VisitBench	.352	+.091	.339	+.064	.524	+.235
ScienceQA	.258	+.075	.304	+.115	.530	+.223
TextVQA	.389	+.092	.213	−.012	.600	+.183
WIT	.164	−.242	.400	+.021	.294	−.109
MM-Vet	.260	−.195	.296	−.108	.233	−.194
AesBench	.402	−.077	.353	−.153	.256	−.208

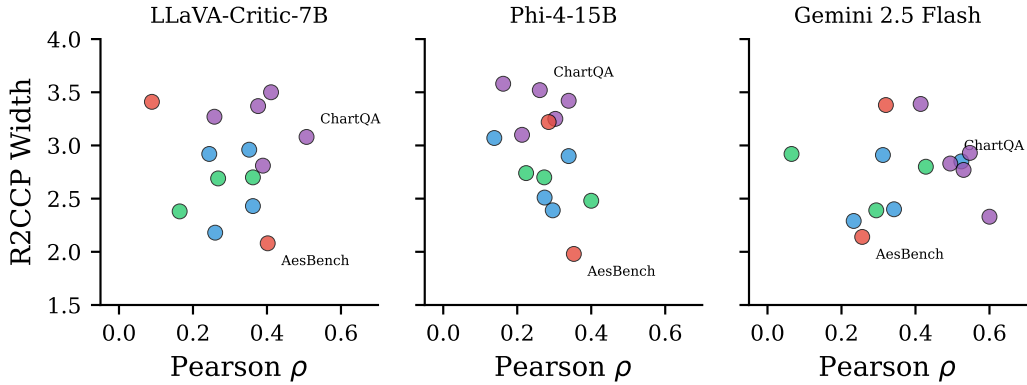


Figure 4: Ranking-scoring decoupling across datasets. High correlation does not imply reliable absolute scores.

Table 18: Conditional coverage by ground truth score.

GT	N	Cov. (raw)	Cov. (adj.)	Width (raw)	Width (adj.)	Bias
1	334	53.0%	88.9%	3.14	3.75	+1.67
2	360	86.9%	100%	3.18	3.79	+1.13
3	637	99.7%	100%	3.09	3.75	+0.77
4	880	98.2%	100%	2.99	3.65	+0.11
5	648	92.4%	99.2%	2.92	3.58	-0.80

Table 19: Naive Split CP vs. R2CCP per dataset (raw width, LLaVA-Critic-7B).

Dataset	Naive	R2CCP	Δ
AesBench	2.53	2.08	-0.45
MM-Vet	3.39	2.18	-1.21
WIT	2.92	2.38	-0.54
COCO	2.86	2.43	-0.43
Mind2Web	3.13	2.69	-0.44
Conc. Cap.	3.11	2.70	-0.41
TextVQA	3.19	2.81	-0.38
LLaVA-B.	3.28	2.92	-0.36
VisitBench	3.37	2.96	-0.41
ChartQA	3.49	3.08	-0.41
ScienceQA	3.71	3.27	-0.44
MathVista	3.64	3.37	-0.27
DiffusionDB	3.56	3.41	-0.15
Infograph.	3.67	3.50	-0.17
Average	3.27	2.84	-0.42

Table 20: Effect of CoT prompting on LLaVA-Critic-7B.

Metric	Generic	CoT
Pearson ρ	.380	.402 (+5.8%)
Spearman ρ_s	.314	.356 (+13.4%)
Kendall τ	.270	.300 (+11.1%)
Exact accuracy	33.9%	32.2%
MAE	1.058	1.031
Overconf. (>0.99)	51.4%	25.6% (-50%)
Overconf. (>0.999)	33.2%	6.6% (-80%)

Table 21: Midpoint vs raw score.

Metric	Raw	Midpoint	Adj. Mid
Pearson	.412	.396	.277
Spearman	.367	.359	.272
MAE	1.022	.992	1.056
Accuracy	32.6%	—	28.1%

Table 22: R2CCP stability as a function of number of seeds (LLaVA-Critic-7B, MLLM-Judge). Results stabilize by 5–10 seeds.

Seeds	Coverage	$\pm$ std	Width	$\pm$ std
5	.901	.016	3.053	.087
10	.900	.016	3.049	.097
15	.903	.014	3.064	.084
20	.902	.014	3.058	.085
25	.902	.014	3.056	.079
30	.902	.013	3.058	.074

Table 23: R2CCP robustness to calibration/test split ratio (LLaVA-Critic-7B, MLLM-Judge, 10 seeds).

Split	Cal N	Test N	Coverage	Width
70/30	4,001	1,715	.900 $\pm$.008	3.05 $\pm$.04
60/40	3,430	2,286	.898 $\pm$.017	3.05 $\pm$.09
50/50	2,858	2,858	.900 $\pm$.016	3.05 $\pm$.10
40/60	2,286	3,430	.896 $\pm$.020	3.06 $\pm$.15
30/70	1,715	4,001	.887 $\pm$.015	2.98 $\pm$.10

Table 24: CP coverage (boundary-adjusted) by error magnitude across all 3 judges (R2CCP, 10 seeds).

Error	LLaVA	Phi-4	Gemini
Exact	100%	99.8%	99.8%
± 1	99.9%	99.9%	99.8%
± 2	99.4%	99.5%	99.0%
± 3	91.5%	86.3%	96.2%
± 4	43.3%	67.4%	68.8%

Table 25: Judge comparison on Polaris. LLaVA-Critic (7B, evaluation-specialized) vastly outperforms Phi-4 (15B, general-purpose) on captioning evaluation.

	LLaVA-Critic 7B	Phi-4 15B (think)	Phi-4 15B (no-think)
Samples	8,726	924	8,726
Pearson	.906	.128	.036
Accuracy	80.9%	26.1%	20.0%
± 1 Acc.	95.4%	60.5%	37.5%
R2CCP Width	0.68	3.60	2.96